Diffusion Model

An AI model that learns to generate Image or video by turning random noise into meaningful content.

Noise: Random values containing no meaning (Pixels).

Example: Imagine opening a T.V with no signal, we see black and white dot appear on the screen.

Image Generation

Example: Tell AI, to generate a picture of white cat sitting on sofa.

How it will work:

1. Start with noise.
2. See the prompt and remove the noise step by step and gradually form the shape.
3. Then produce final image.

During training it learns patterns form many images like cat ears, eyes, fur… etc.

We only see the final result not the noise.

Video Generation

Frame: One single image inside a video.

Example: Imagine a person walking and you take picture of the person at one moment, then another picture a tiny moment later at different position. Those individual picture are called frame.

How many frames are in 1 sec. it depends on video FPS.

24 FPS ----- 24 Frames Per Second.

30 FPS ----- 30 Frames Per Second.

$$\text{Total Frames} = \text{FPS} \times \text{Duration}.$$

OpenAI has a video-generation tool called Sora, designed to create videos from text prompts.

Example: Tell AI, to generate a video of white cat running across the garden.

1. Understand the prompt.
2. Generate video by generating sequence of characters.
3. The model tries to maintain cat's appearance, so that its movement looks natural.

However, OpenAI discontinued the Sora app and website on April 26, 2026, and its API was discontinued on September 24, 2026. So, you cannot currently use Sora as OpenAI's video-generation option.

But you can try other options like,

1. Google Veo
2. Runway
3. Google Flow
4. Gemini
5. Canva